

AGNEL INSTITUTE OF TECHNOLOGY AND DESIGN

AND GATE:

```
import numpy as np
def mc_culloch_pit(x, w, threshold):
    weighted_sum = np.dot(x, w)
    if weighted_sum >= threshold:
        return 1
    else :
        return 0

X=np.array([[0,0], [0,1], [1,0], [1,1]])
w=[]
print(f"Enter inputs for weights: ")
for i in range(0,2):
    print("W", i, " : ", end="")
    element = int(input())
    w.append(element)

print(f"Enter inputs for threshold : ")
threshold = int(input())

for x in X:
    output = mc_culloch_pit(x, w, threshold)
    print(f"Input : {x}, Output : {output}")
```

Output:

```
Enter inputs for weights:
W 0 : 1
W 1 : 1
Enter inputs for threshold :
2
Input : [0 0], Output : 0
Input : [0 1], Output : 0
Input : [1 0], Output : 0
Input : [1 1], Output : 1
```

NOR GATE

```
import numpy as np
def mc_culloch_pit(x, w, threshold):
    weighted_sum = np.dot(x, w)
    if weighted_sum >= threshold:
        return 1
    else :
```

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```
return 0
```

```
X=np.array([[0,0], [0,1], [1,0], [1,1]])
w=[]
print(f"Enter inputs for weights: ")
for i in range(0,2):
    print("W", i, " : ", end="")
    element = int(input())
    w.append(element)

print(f"Enter inputs for threshold : ")
threshold = int(input())

for x in X:
    output = mc_culloch_pit(x, w, threshold)
    print(f"Input : {x}, Output : {output}")
```

Output:

```
Enter inputs for weights:
W 0  : -1
W 1  : -1
Enter inputs for threshold :
0
Input : [0 0], Output : 1
Input : [0 1], Output : 0
Input : [1 0], Output : 0
Input : [1 1], Output : 0
```

OR GATE:

```
import numpy as np
def mc_culloch_pit(x, w, threshold):
    weighted_sum = np.dot(x, w)
    if weighted_sum >= threshold:
        return 1
    else :
        return 0
```

```
X=np.array([[0,0], [0,1], [1,0], [1,1]])
w=[]
```

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```
print(f"Enter inputs for weights: ")
for i in range(0,2):
    print("W", i, " : ", end="")
    element = int(input())
    w.append(element)

print(f"Enter inputs for threshold : ")
threshold = int(input())

for x in X:
    output = mc_culloch_pit(x, w, threshold)
    print(f"Input : {x}, Output : {output}")
```

Output:

```
Enter inputs for weights:
W 0 : 1
W 1 : 1
Enter inputs for threshold :
1
Input : [0 0], Output : 0
Input : [0 1], Output : 1
Input : [1 0], Output : 1
Input : [1 1], Output : 1
```

NAND:

```
import numpy as np
def mc_culloch_pit(x, w, threshold):
    weighted_sum = np.dot(x, w)
    if weighted_sum >= threshold:
        return 0
    else :
        return 1
```

```
X=np.array([[0,0], [0,1], [1,0], [1,1]])
w=[]
print(f"Enter inputs for weights : ")
for i in range(0,2):
    element = int(input())
    w.append(element)
print(f"Enter inputs for threshold : ")
threshold = int(input())
for x in X:
```

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```
output = mc_culloch_pit(x, w, threshold)
print(f"Input : {x}, Output : {output}")
```

Output:

```
Enter inputs for weights:
W 0 : 1
W 1 : 1
Enter inputs for threshold :
2
Input : [0 0], Output : 0
Input : [0 1], Output : 0
Input : [1 0], Output : 0
Input : [1 1], Output : 1
```

x1 AND(NOTx2) GATE:

```
import numpy as np
def mc_culloch_pit(x, w, threshold):
    weighted_sum = np.dot(x, w)
    if weighted_sum >= threshold:
        return 1
    else :
        return 0

X = np.array([[0,0], [0,1], [1,0], [1,1]])
w = []
print(f"Enter inputs for weights: ")
for i in range(0,2):
    print("W", i, " : ", end="")
    element = int(input())
    w.append(element)

print(f"Enter inputs for threshold : ")
threshold = int(input())

for x in X:
    output = mc_culloch_pit(x, w, threshold)
    print(f"Input : {x}, Output : {output}")
```

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Output:

```
Enter inputs for weights:
W 0 : 1
W 1 : -1
Enter inputs for threshold :
1
Input : [0 0], Output : 0
Input : [0 1], Output : 0
Input : [1 0], Output : 1
Input : [1 1], Output : 0
```

NOT GATE:

```
import numpy as np
def mc_culloch_pit(x, w, threshold):
    weighted_sum = np.dot(x, w)
    if weighted_sum < threshold:
        return 1
    else :
        return 0

X = np.array([1,0])
w=[]
print(f"Enter inputs for weights: ")
for i in range(0,1):
    print("W", i, " : ", end="")
    element = int(input())
    w.append(element)

print(f"Enter inputs for threshold : ")
threshold = int(input())

for x in X:
    output = mc_culloch_pit(x, w, threshold)
    print(f"Input : {x}, Output : {output}")
```

Output:

```
Enter inputs for weights:
W 0 : 1
Enter inputs for threshold :
1
Input : 1, Output : 0
Input : 0, Output : 1
```